

Sample size calculations for prognostic model research

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Many thanks to Gary Collins, Kym Snell, Joie Ensor,
Maarten Van Smeden, Frank Harrell, Carl Moons, Thomas
Debray, Lucinda Archer, Glen Martin & others

Outline

- **Sample size for model development**
 - instability with small sample sizes
 - recent proposals
- **Sample size for model validation**
 - outline of new proposals

Part 1

SAMPLE SIZE FOR PROGNOSTIC MODEL DEVELOPMENT

Why is sample size important?

Model development

- We want to have a large enough sample size to develop a model that provides accurate risk predictions in new individuals from target population
- Many (most?) models do not perform well in new data

Why?

Development studies often

- have small sample sizes (predictor effects have large imprecision)
- use misleading rule of thumb ('10 events per variable')
- are overfitted (predictor effects are too extreme)
- lack internal validation & optimism-adjustment (shrinkage)

Small sample size = miscalibration

Calibration

- Agreement between predicted risks (from the model) and observed risks (in the data)
- Ideally assessed in data independent from development dataset
- Can be examined using calibration plots & calibration curves

Van Calster *et al. BMC Medicine* (2019) 17:230
<https://doi.org/10.1186/s12916-019-1466-7>

BMC Medicine

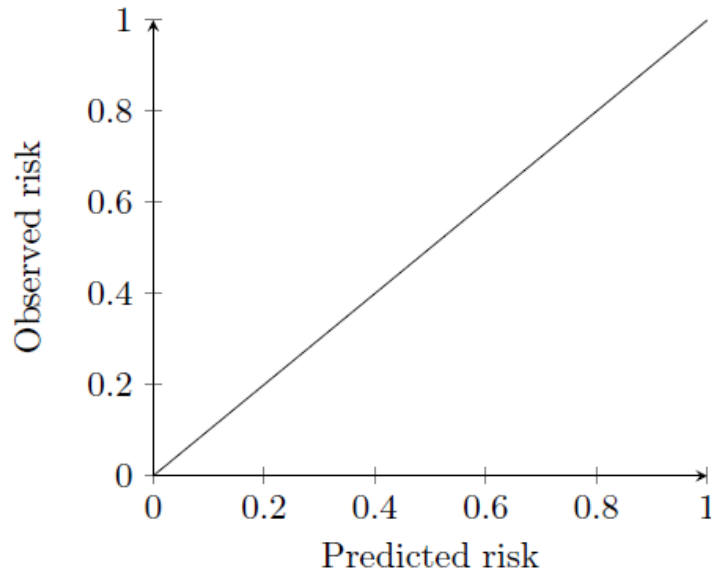
OPINION

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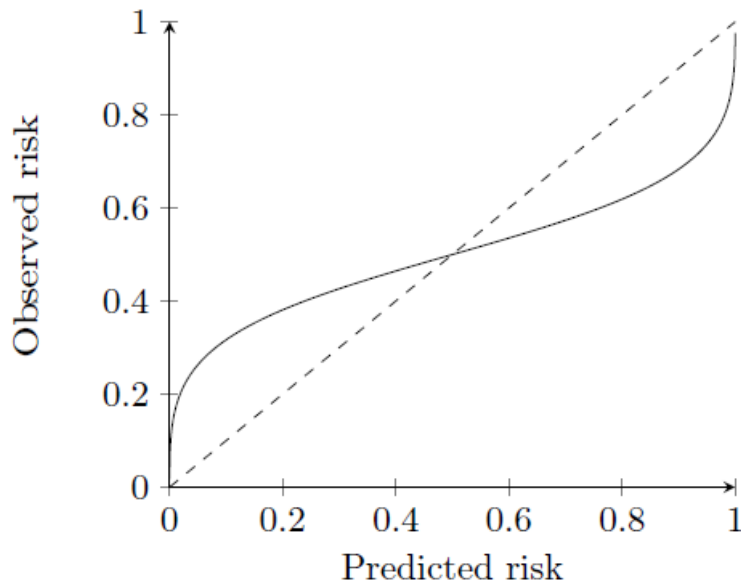
Calibration: the Achilles heel of predictive analytics



Ben Van Calster^{1,2,6*} , David J. McLernon^{3,6} , Maarten van Smeden^{2,4,6} , Laure Wynants^{1,5}, Ewout W. Steyerberg^{2,6} 
On behalf of Topic Group 'Evaluating diagnostic tests and prediction models' of the STRATOS initiative⁶



- IDEAL MODEL:
- perfect calibration
 - calibration slope = 1



- OVERFITTED MODEL:
- predicted risks too extreme
 - calibration slope < 1
 - may lead to harm!

Penalisation & shrinkage to address overfitting

- Modern methods aim to resolve overfitting
 - target calibrated predicted risks in new data
- Penalised regression:
 - lasso, elastic net, ridge regression
- Standard (unpenalised) regression
 - followed by uniform (global) shrinkage
- With shrinkage & penalty terms estimated using bootstrapping or cross-validation

So shrinkage resolves overfitting – right?

- **No – we should still aim for overfitting to be small**
- Models are more robust when shrinkage required is close to 1
- Van Houwelingen (2001) notes that:

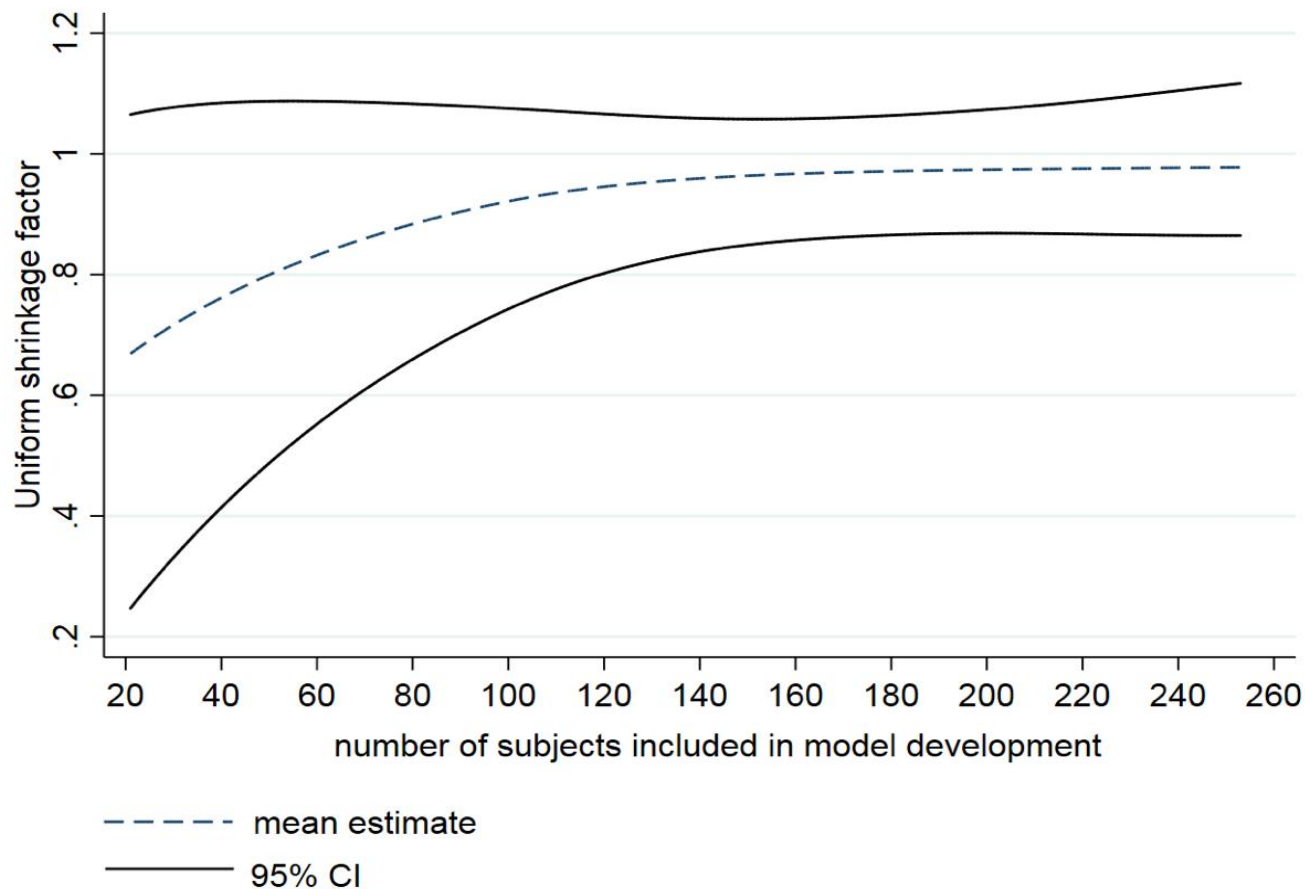
“... shrinkage works on the average but may fail in the particular unique problem on which the statistician is working”

Van Houwelingen JC. Shrinkage and penalized likelihood as methods to improve predictive accuracy. *Statistica Neerlandica*. 2001;55:17-34

- **Shrinkage (penalty / tuning) factors are hard to estimate**
- **Often large uncertainty in the correct value to use**

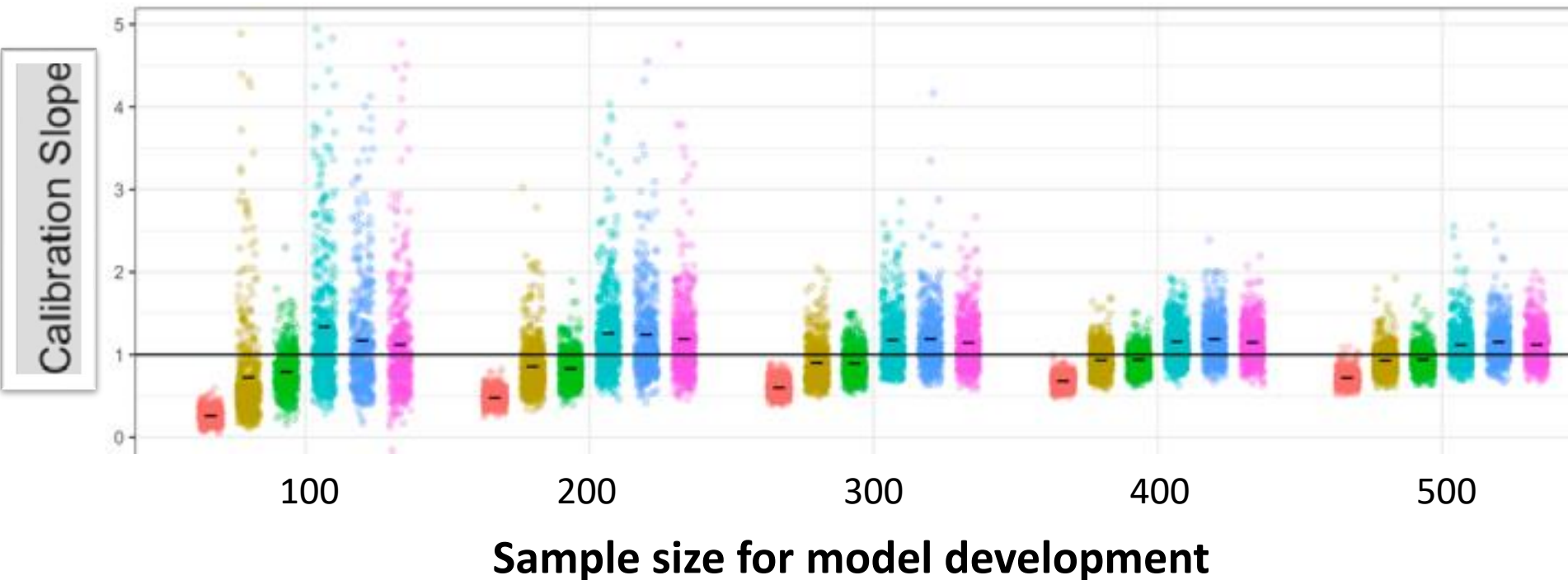
Illustration (Riley et al., J Clin Epi (2020))

- **Predicting 1-year BP outcome in hypertensive patients**
- Linear regression, 7 predictors, R-squared about 50%
- What shrinkage is needed after using 20 to 253 participants?



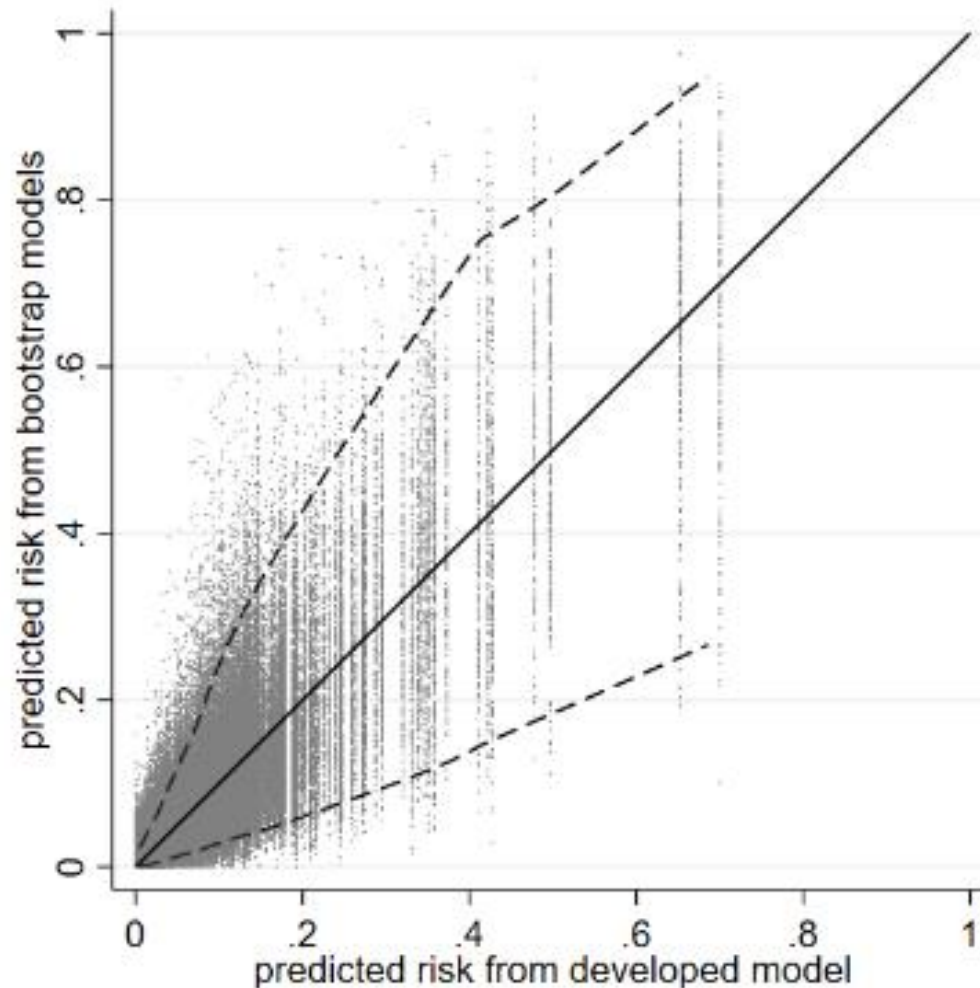
Sim study (Riley et al., 2020): 20 predictors, outcome incidence 50%

Calibration slope in large validation dataset



- Maximum likelihood
- Bootstrap shrinkage
- Elasticnet
- Heuristic shrinkage
- Ridge
- Lasso

Instability plot of individual risks across bootstrap lasso models
- 752 patients, 53 events, 27 predictors (7 + 20 noise), and EPP of 2
(Riley & Collins, in preparation)



So shrinkage is important but not guaranteed to solve overfitting

Article



Regression shrinkage methods for clinical prediction models do not guarantee improved performance: Simulation study

Statistical Methods in Medical Research
0(0) 1–13

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DOI: 10.1177/0962280220921415

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Ben Van Calster^{1,2} , Maarten van Smeden^{2,3} ,
Bavo De Cock^{1,4} and Ewout W Steyerberg²

“The results imply that shrinkage methods do not solve problems associated with small sample size or low number of events per variable”

Also see:

Riley RD, Snell KIE, et al. Penalisation and shrinkage methods produced unreliable clinical prediction models especially when sample size was small.

J Clin Epi 2021; 132:88-96

Our Sample Size Proposal

- Target: more reliable (stable) models
- Proposals published in BMJ and Stat Med (refs at the end) since 2019

PROPOSAL:

- Calculate sample size that is needed to
 - minimise potential overfitting
 - estimate overall risk precisely
- **Requires calculations for multiple criterion**

Minimum sample size for binary & time-to-event outcomes

- Three criteria to meet:
 - (i) A uniform shrinkage factor, $S \geq 0.9$
(e.g. target < 10% overfitting)
 - (i) A small difference in Nagelkerke's R_{app}^2 and R_{adj}^2
(e.g. target < 0.05 absolute difference)
 - (iii) A small margin of error in overall risk estimate
(e.g. target < 0.05 absolute error)

Minimum sample size for binary & time-to-event outcomes

- Closed-form formulae for each criteria (so quick!)
- Requires the researcher to pre-specify
 - the number of predictor parameters (p)
 - the desired heuristic shrinkage factor (S)
 - the overall risk in the population
 - model's anticipated Cox-Snell R^2
- Use information from previous studies and models, such as overall risk, predictive performance (R^2 , C)

Minimum sample size for binary & time-to-event outcomes

- Can obtain R^2 from a reported C-statistic for an existing model

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RESEARCH ARTICLE

Statistics
in Medicine WILEY

A note on estimating the Cox-Snell R^2 from a reported C statistic (AUROC) to inform sample size calculations for developing a prediction model with a binary outcome

Richard D. Riley¹  | Ben Van Calster^{2,3}  | Gary S. Collins^{4,5} 

Criteria (i): target shrinkage S of 0.9

n sample size, p predictor parameters,

R^2 Cox-Snell R-squared, LR likelihood ratio statistic

- Binary or time-to-event outcomes (van Houwelingen & Le Cessie)

$$S = 1 - \frac{p}{\text{LR}}$$

$$S = 1 + \frac{p}{n \log \left(1 - \frac{R_{\text{adj}}^2}{S} \right)}$$

- Hence
$$n = \frac{p}{(S-1) \log \left(1 - \frac{R_{\text{adj}}^2}{S} \right)}$$

Cox-Snell R^2 may be small

- For logistic & Cox models the Cox-Snell R^2 has a maximum value less than 1, defined by

$$\max(R^2) = 1 - \exp\left(\frac{2\log L_{null}}{n}\right)$$

- **So be realistic about the potential value of the Cox-Snell R^2**
- For example, for a logistic regression model with
 - an outcome proportion of 50%, $\max(R^2)$ is 0.75
 - an outcome proportion of 5%, $\max(R^2)$ is 0.33
 - an outcome proportion of 1%, $\max(R^2)$ is 0.11

Stata module by Ensor et al: PMSAMPSIZE

Binary outcome example: Cox-Snell R^2 0.2, outcome 50%, p = 30

```
. pmsampsize, type(b) rsquared(0.2) parameters(30) prev(0.5)
```

	Samp_size	Shrinkage	Parameter	Rsq	Max_Rsq	EPP
Criteria 1	1194	.9	30	.2	.75	19.9
Criteria 2	701	.842	30	.2	.75	11.68
Criteria 3	385	.9	30	.2	.75	6.42
Final	1194	.9	30	.2	.75	19.9

Minimum sample size required = 1194, with 597 events

EPP = 19.9

Stata module by Ensor et al: PMSAMPLESIZE

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Final	1194	.9	30	.2	.75	19.9

Minimum sample size required = 1194, with 597 events

EPP = 19.9 (TWICE THE RULE OF 10 EVENT EVENTS PER VARIABLE)

Stata module by Ensor et al: PMSAMPSIZE

Binary outcome example: Cox-Snell R^2 0.5, outcome 50%, p = 30

```
. pmsampsize, type(b) rsquared(0.5) parameters(30) prev(0.5)
```

	Samp_size	Shrinkage	Parameter	Rsqr	Max_Rsq	EPP
Criteria 1	370	.9	30	.5	.75	6.17
Criteria 2	556	.93	30	.5	.75	9.27
Criteria 3	385	.93	30	.5	.75	6.42
Final	556	.93	30	.5	.75	9.27

Minimum sample size required = 556, with 278 events

EPP = 9.27

Minimum required!

- **The sample size that meets all criteria is the MINIMUM required**
- **Why minimum?** Other criteria may be important
e.g. ensure precise estimation of key predictor effects
- Might examine a few key predictors (strong prognostic value)
- **Will raise the sample size further but give even more stability**

A further criterion

- Van Smeden et al. propose an empirically based sample size equation **to ensure small mean absolute prediction error**

<https://mvansmeden.shinyapps.io/BeyondEPV/>

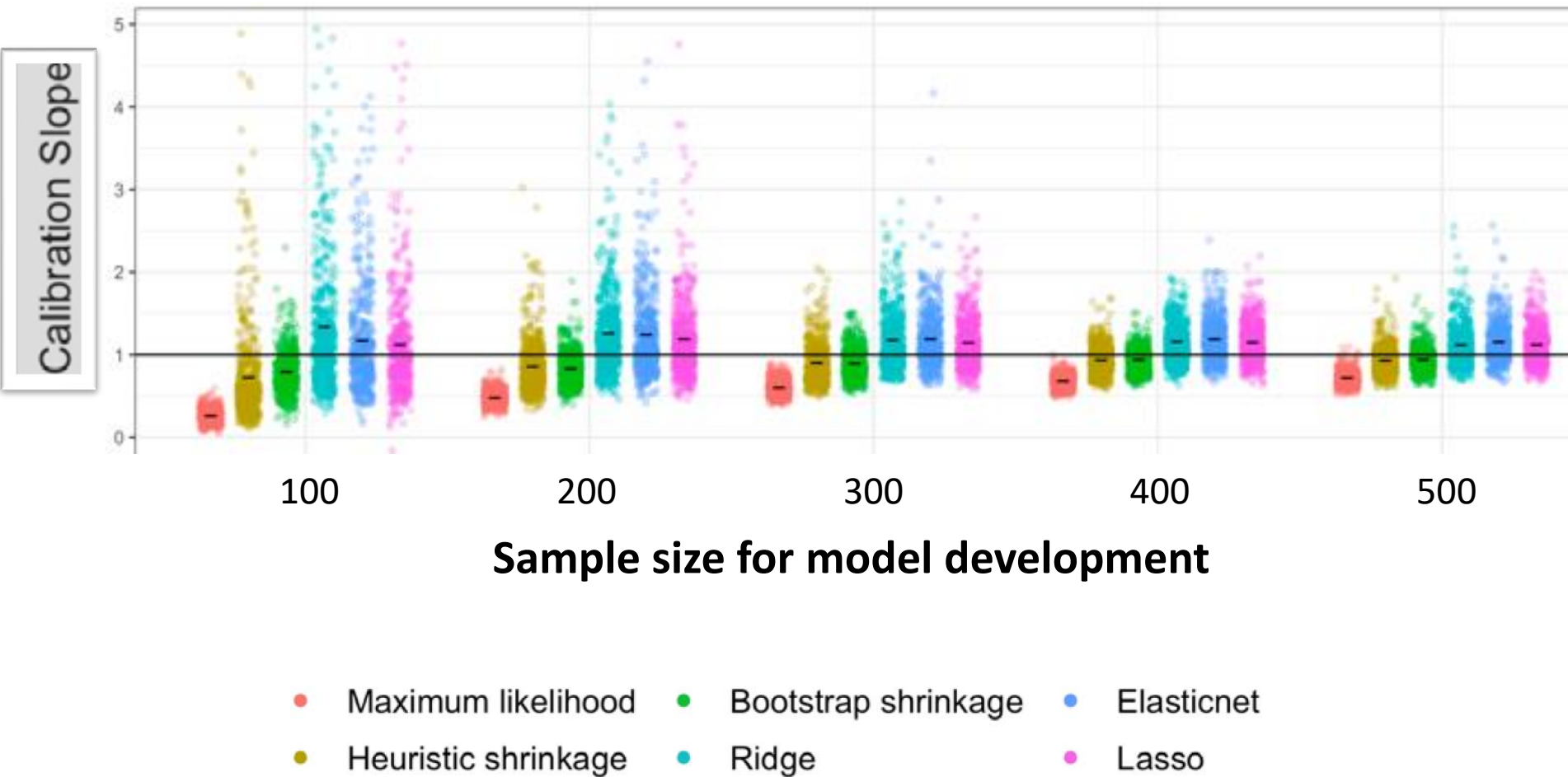
- This is another criteria to consider for binary outcomes

van Smeden M, Moons KG, de Groot JA, Collins GS, Altman DG, Eijkemans MJ, et al. Sample size for binary logistic prediction models: Beyond events per variable criteria. Stat Methods Med Res (2019)

- **Our proposal focuses on when we want to include all our predictors (p) into the model**
- What about variable selection situations?
- Lasso & elastic net allow this – **but stability in new data a concern**
- **Hence, still adhere to our sample size guidance**

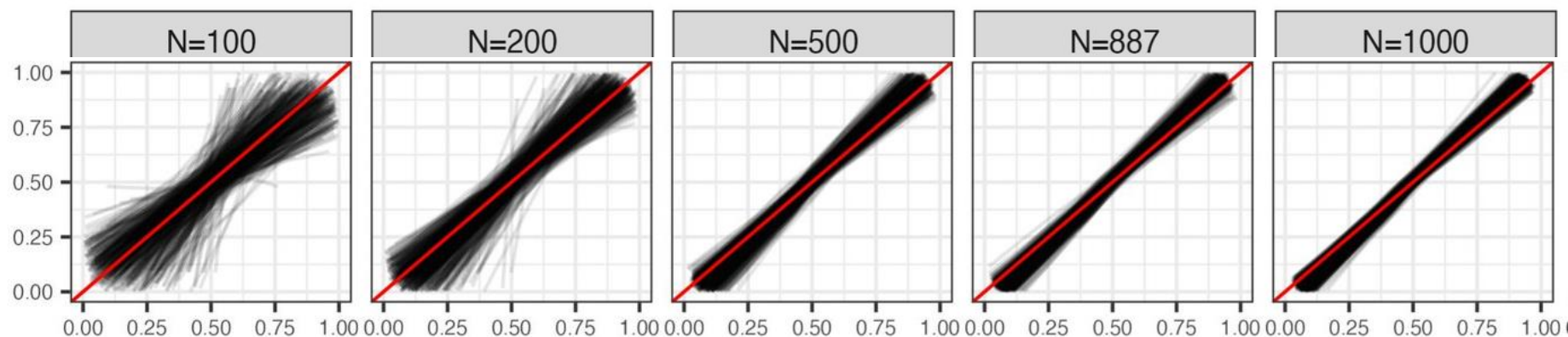
Sim study (Riley et al., 2020): 20 predictors, outcome incidence 50%

Calibration slope in large validation dataset



Calibration curves in the true population after using elastic net

- each black curve is the true calibration (in large validation data) of an EL model generated in a random sample of size N

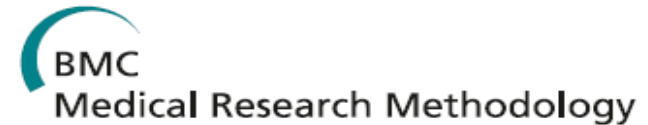


Instability decreases

(Riley & Collins, in preparation)

Machine learning is our saviour?

van der Ploeg *et al.* *BMC Medical Research Methodology* 2014, **14**:137
<http://www.biomedcentral.com/1471-2288/14/137>



RESEARCH ARTICLE

Open Access

Modern modelling techniques are data hungry: a simulation study for predicting dichotomous endpoints

Tjeerd van der Ploeg^{1,3*}, Peter C Austin² and Ewout W Steyerberg³

“Modern modelling techniques such as Support Vector Machines, Neural Networks and Random Forests **may need over 10 times as many EPV to achieve a stable C-statistic and a small optimism than classical modelling techniques such as logistic regression.** This implies that such modern techniques should only be used in medical prediction problems if very large data sets are available”

Machine learning is our saviour?

- Much larger sample sizes are required
 - Why? Many more parameters considered
 - Nonlinearity, interactions etc
- May need over 10 times as many events per variable to achieve stable discrimination, compared to logistic regression
- Random forests particularly unstable even at EPV=200

van der Ploeg et al. BMC Med Res Methodol 2014

Using meaningful prior information may help ...

- Šinkovec, H., Heinze, G., Blagus, R. *et al.* To tune or not to tune, a case study of ridge logistic regression in small or sparse datasets. *BMC Med Res Methodol* **21**, 199 (2021).

“Applying tuned ridge regression in small or sparse datasets is problematic as it results in unstable coefficients and predictions. In contrast, determining the degree of shrinkage according to some meaningful prior assumptions about true effects has the potential to reduce bias and stabilize the estimates”

Part 3

SAMPLE SIZE FOR MODEL VALIDATION

“I always think part of success is being able to replicate results seeing if you bring it into different situations with similar results”

Which statistics legend said this?

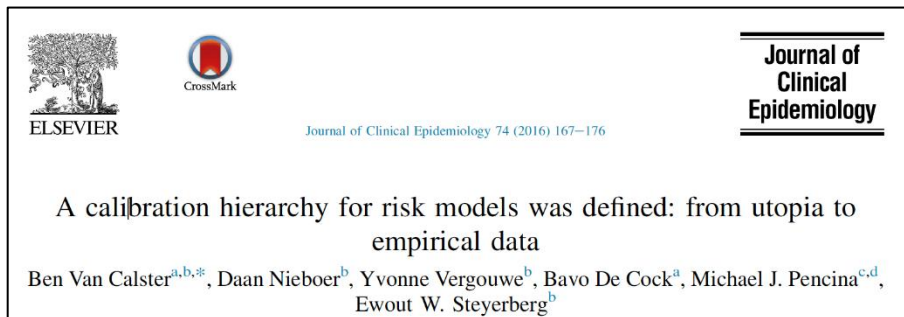
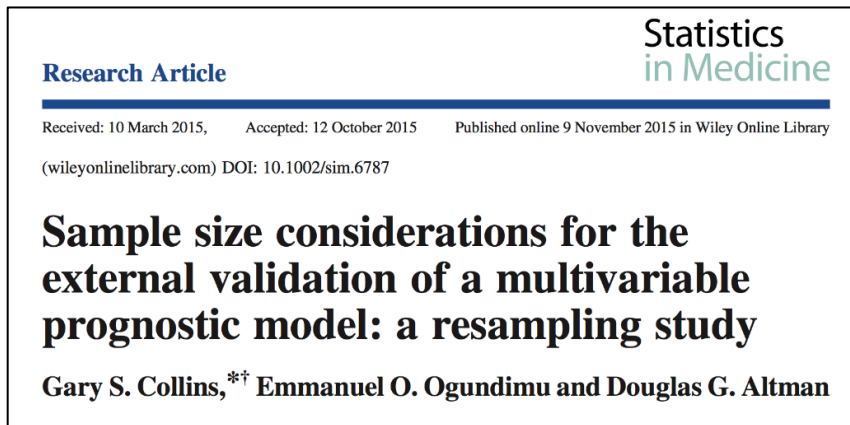
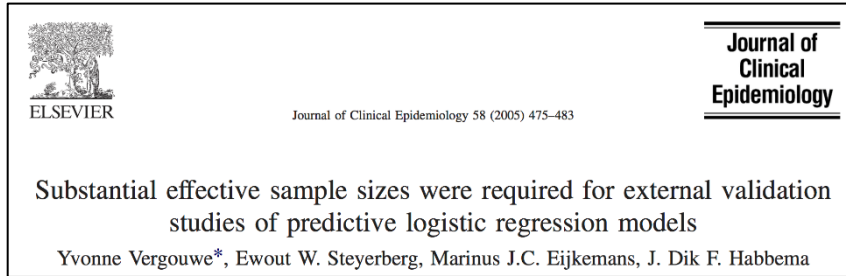
“I always think part of success is being able to replicate results seeing if you bring it into different situations with similar results”

Robert Downey, Jr. (Iron Man)

External validation

- Focus of validation is to evaluate model performance in new data
- If it performs well in new data, who cares how it was developed?
- We want an adequate sample size to accurately and precisely assess
 - model discrimination
 - model calibration(perhaps also clinical utility, e.g. net benefit)

Binary and time-to-event outcomes



Rules of thumb proposed for sample size:

- Minimum of 100 events and 100 non-events
- For calibration, might even want > 200 events
- Can we produce a more tailored approach?

Precision in performance measures

Let us focus on validation of a prediction model for binary outcome

EXISTING MODEL: $\text{logit}(\hat{p}_i) = \hat{\alpha} + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \dots = LP_i$

- Consider a **calibration model** to validate fit of predictions in new data

$$\text{logit}(p_i) = \gamma + \beta(LP_i)$$

- What influences the precision of the calibration slope (β)?
 - Sample size
 - Distribution of LP_i
 - Outcome proportion (number of events)
 - Values of γ and β (0 and 1 indicate 'perfect' calibration)

Precision in performance measures

Let us focus on validation of a prediction model for binary outcome

EXISTING MODEL: $\text{logit}(\hat{p}_i) = \hat{\alpha} + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 X_{3i} + \dots = LP_i$

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 - Values of γ and β (0 and 1 indicate 'perfect' calibration)

Simulation-based sample size (Snell et al., J Clin Epi 2021)

Calibration model: $\text{logit}(p) = \gamma + \beta(LP)$

1. Specify distribution of LP_i
2. Specify parameters γ and β
3. Specify target precision for each performance measure
4. Specify starting sample size & number of repetitions to use for simulation
5. Generate LP_i and outcome values for each participant
6. Calculate performance measures and CIs
7. Repeat for specified number of repetitions
8. Calculate estimates of precision
9. Adjust sample size and repeat 5-8 until target precision is achieved for all performance measures

Example: Deep Vein Thrombosis (diagnostic) prediction model

- From development article:

$LP_i \sim N(-1.75, 1.47^2)$ Outcome prevalence 22%

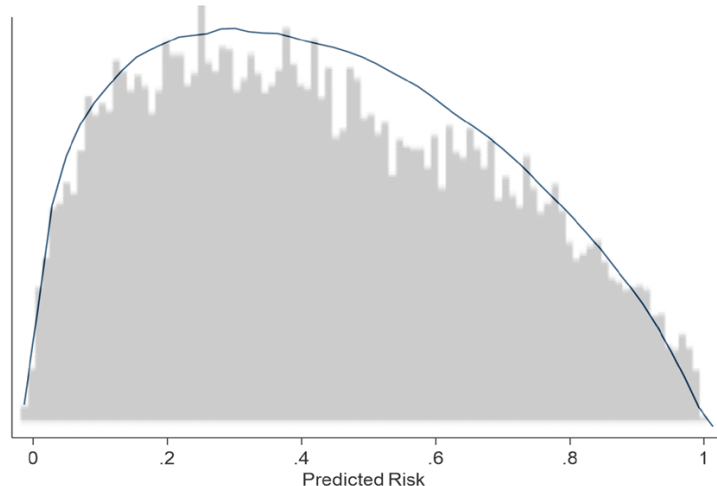
Performance Measure	Targeted 95% CI width	Sample size (events) required to achieve CI width
C-statistic	0.1	385 (85)
Calibration slope	0.2	2430 (531)
O/E	0.2	1379 (302)

Closed-form solutions (Riley et al., 2021)

- In a further paper we use Fisher's information to derive analytic (closed-form) solutions for the standard errors of
 - Calibration slope
 - O/E
 - C-statistic
 - Net benefit
- Rearrange to give sample size required to achieve target standard error. User must specify:
 - (i) target standard errors (confidence interval widths)
 - (ii) the anticipated outcome proportion
 - (iii) the prediction model's anticipated (mis)calibration and variance of LP_i values in the validation population
 - [(iv) potential risk thresholds for clinical decision making]

Closed-form solutions (Riley et al., 2021)

- Stata and R code provided
- How gauge the anticipated distribution of LP_i values?
 - (i) extract information from development study
 - you may have the IPD
 - could ask model developers
 - sometimes mean and variance provided
 - sometimes histogram of risks is provided: example ...



Closed-form solutions (Riley et al., 2021)

- How gauge the anticipated distribution of LP_i values?
 - (ii) under (strong) assumptions of normality with common variance in events and non-events groups, the variance of LP_i values is a function of the C-statistic
 - hence can use reported (optimism-adjusted) C-statistic from development study

Example

- Calculate sample size required to target precise performance estimates in an external validation study of a DVT prediction model, assuming $LP \sim N(-1.75, 1.47^2)$ and model is well calibrated, with outcome proportion of 0.22

Performance Measure	Assumed value	Targeted 95% CI width	Results using simulation-based approach	Results using closed-form approach
			Sample size (events) required to achieve CI width	Sample size (events) required to achieve CI width
C-statistic	0.82	0.1	385 (85)	400 (88)
Calibration slope	1	0.2	2430 (531)	2407 (530)
O/E	1	0.2	1379 (302)	1364 (300)

- Very similar findings using simulation or closed-form
- Simulation approach slower but more flexible

Survival outcomes

Riley RD, Collins GS, Ensor J, et al. Minimum sample size calculations for external validation of a clinical prediction model with a time-to-event outcome. Stat Med 2022

Closed-form solutions not possible

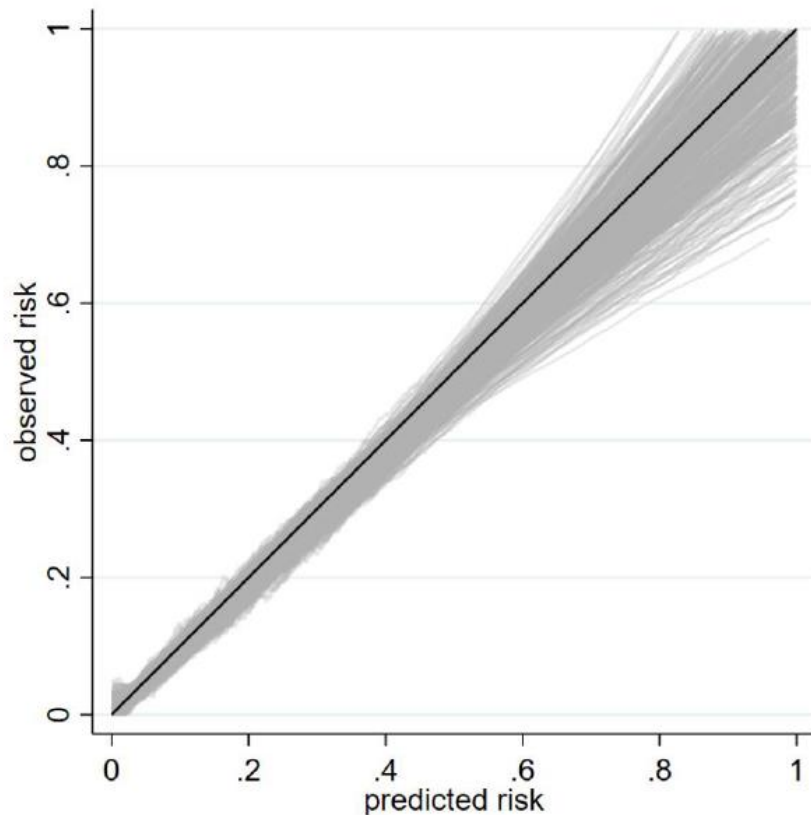
Simulation-based approach needed, but complex because:

- Need to specify survival distribution
- Need to specify censoring distribution
- Need to focus on one (or a few) particular time-points
- Need to use pseudo-values to overcome 'censoring' (true value not observed by particular times)

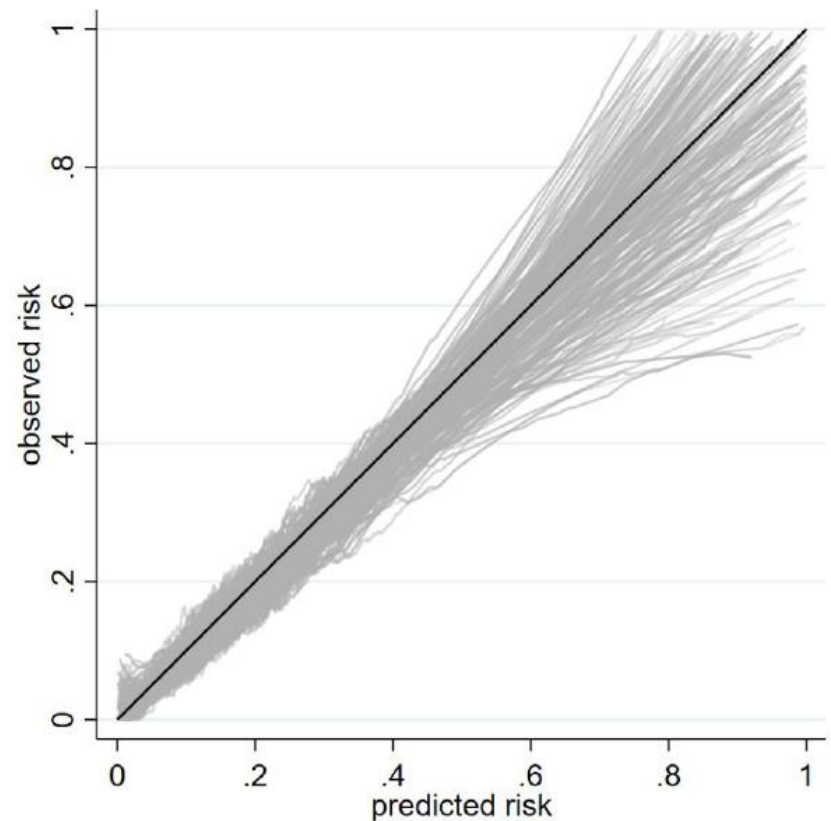
How do we define precise?

Plotting empirical distribution of calibration curves helps for particular sample sizes

(a) 14250 individuals



(b) 3600 individuals



Thank You ... & a final thought

*“Prediction is not just one of the things your brain does. It is the primary function of the neo-cortex, and the **foundation of intelligence**”*

Jeff Hawkins

- Intelligence includes using sound statistical principles
- Encourage this in your prediction research projects

FURTHER INFORMATION

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DEVELOPMENT

- Van Smeden et al. Sample size for binary logistic prediction models: Beyond events per variable criteria
- Van Smeden et al. No rationale for 1 variable per 10 events criterion for binary logistic regression analysis
- Riley RD, Snell KI, Ensor J, et al. Minimum sample size for developing a multivariable prediction model: Part II - binary and time-to-event outcomes. *Stat Med*. 2019;38(7):1276-96.
- Riley RD, Snell KIE, Ensor J, et al. Minimum sample size for developing a multivariable prediction model: Part I - Continuous outcomes. *Stat Med*. 2019;38(7):1262-75
- Riley RD, Ensor J, et al. Calculating the sample size required for developing a clinical prediction model. *BMJ* 2020;368:m441.
- Riley RD, Van Calster B, Collins GS. A note on estimating the Cox-Snell R(2) from a reported C statistic (AUROC) to inform sample size calculations for developing a prediction model with a binary outcome. *Stat Med* 2021;40(4):859-64.
- Van Calster et al. Adaptive sample size determination for the development of clinical prediction models
- Van der Ploeg et al. Modern Modelling Techniques Are Data Hungry: A Simulation Study for Predicting Dichotomous Endpoints
- Riley et al. Penalization and shrinkage methods produced unreliable clinical prediction models especially when sample size was small
- Van Calster et al. Regression shrinkage methods for clinical prediction models do not guarantee improved performance

VALIDATION

- Archer L, Snell KIE, Ensor J, et al. 2020. Minimum sample size for external validation of a clinical prediction model with a continuous outcome. *Stat Med* 2021; 40: 133-146
- Riley R, Debray TP, Collins GS et al. Minimum sample size for external validation of a clinical prediction model with a binary outcome. *Stat Med* 2021;40:4230-4251
- Riley RD, Collins GS, Ensor J, et al. Minimum sample size calculations for external validation of a clinical prediction model with a time-to-event outcome. *Stat Med* 2022; 17: 1280-1295
- Pavlou et al. Estimation of required sample size for external validation of risk models for binary outcomes. *Stat Med (in-press)*